



REAL ESTATE FINANCE AND INVESTMENTS

RISKS AND OPPORTUNITIES

Chapter 9

The Use and Selection of Cap Rates

Basic Valuation

- Multiple analysis
 - Estimate the value of a property by multiplying next year's NOI
- Cap rates are the inverse of the multiple
 - Stabilized NOI divided by property value (purchase price)
 - Historic linkage to bond market
 - Relevant when NOI is projected to be flat or growing smoothly

Figure 9.1

Capitalization Rate, Income Multiple and Valuation Equations

Cap Rate = Stabilized NOI / Value = 1 / Multiple

Multiple = Value / Stabilized NOI = 1 / Cap Rate

Value = Multiple * Stabilized NOI = Stabilized NOI / Cap Rate

Example: an "8 cap" = .08 = 8.00% = 12.5x Multiple

Figure 9.2

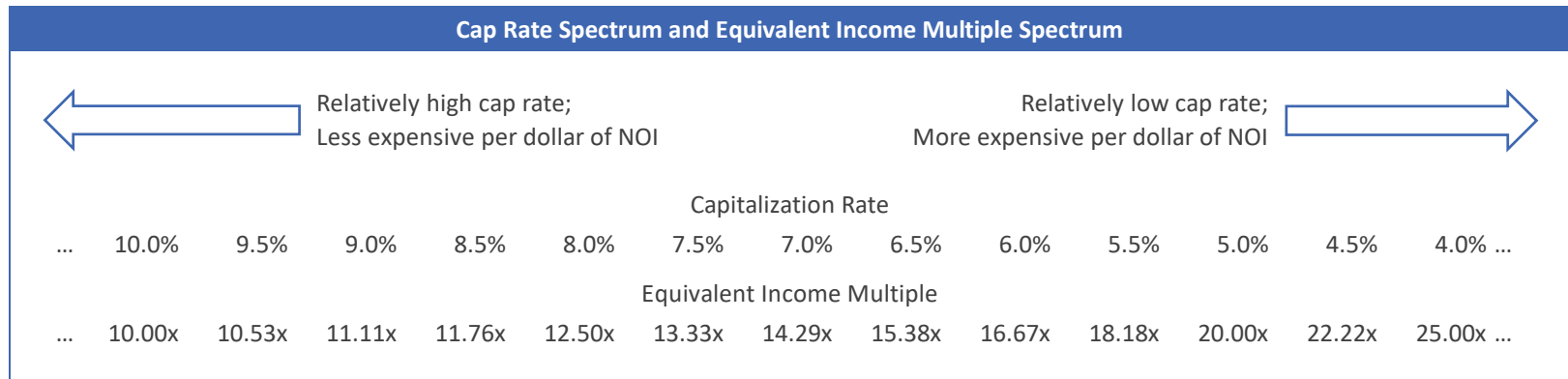


Figure 9.3

Felicitas Tower Cap Rate and Multiple Valuation

Assumptions

Yield on 10-Year T Note	1.60%
Risk Premium	300 basis points (bp)
Cap Rate	$1.60\% + 300 \text{ bp} = 4.60\%$
Multiple	$1 / .046 = 21.7x$
NOI	\$3 MM

Valuation

With Cap Rate	$\$3 \text{ MM} / .046 = \65.2 MM
With Multiple	$\$3 \text{ MM} * 21.7 = \65.2 MM

Replacement Cost

- Indicative of new product entry
- Building cost vs. land cost

Gordon Model: Cap Rate Estimation

- Value = Stabilized NOI / (Discount Rate - Growth Rate) = Stabilized NOI / Cap
- Cap = $(r - g)$
- Model only works if $r > g$, and if g is constant in perpetuity
- NOI and NOI growth increase value
- Risk (the Discount Rate) decreases value

Figure 9.4

Gordon Growth Model for Property Valuation *

Value = NOI / (property discount rate – NOI growth rate) = NOI / Cap Rate

Value = NOI / (r – g) = NOI / Cap Rate

Therefore: Cap Rate = (r – g)

**Assumes Stabilized NOI and constant NOI growth rate for perpetuity.*

Figure 9.5

Gordon Model Cap Rate and Multiple Calculation for Felicitas Tower

Property Discount Rate (r)	4.60%
NOI Growth Rate (g)	1.00%
Cap Rate Calculation (r – g)	$.046 - .01 = .036 = 3.60\%$
Multiple Calculation	$1 / (.046 - .01) = 27.8x$

Figure 9.6

Felicitas Tower Gordon Model Income Multiple Valuation With and Without Growth	
Discount Rate	5.60%
Growth Rate	1.00%
Multiple with growth	$1 / (r - g) = 1 / (.056 - .01) = 21.7x$
Multiple without growth	$1 / r = 1 / .056 = 17.9x$
<u>Valuation</u>	
NOI this year	\$3.00 MM
NOI next year	$\$3.00 \text{ MM} * 1.01 = \3.03 MM
Value with growth	$\$3.03 \text{ MM} * 21.7 = \65.8 MM
Value without growth	$\$3.00 \text{ MM} * 17.9 = \53.7 MM

Market Change

- Discount Rate = risk-free rate + risk premium
- Impact of changes in the risk-free rate
- Market change can impact risk premium
 - A look at the past

Figure 9.7

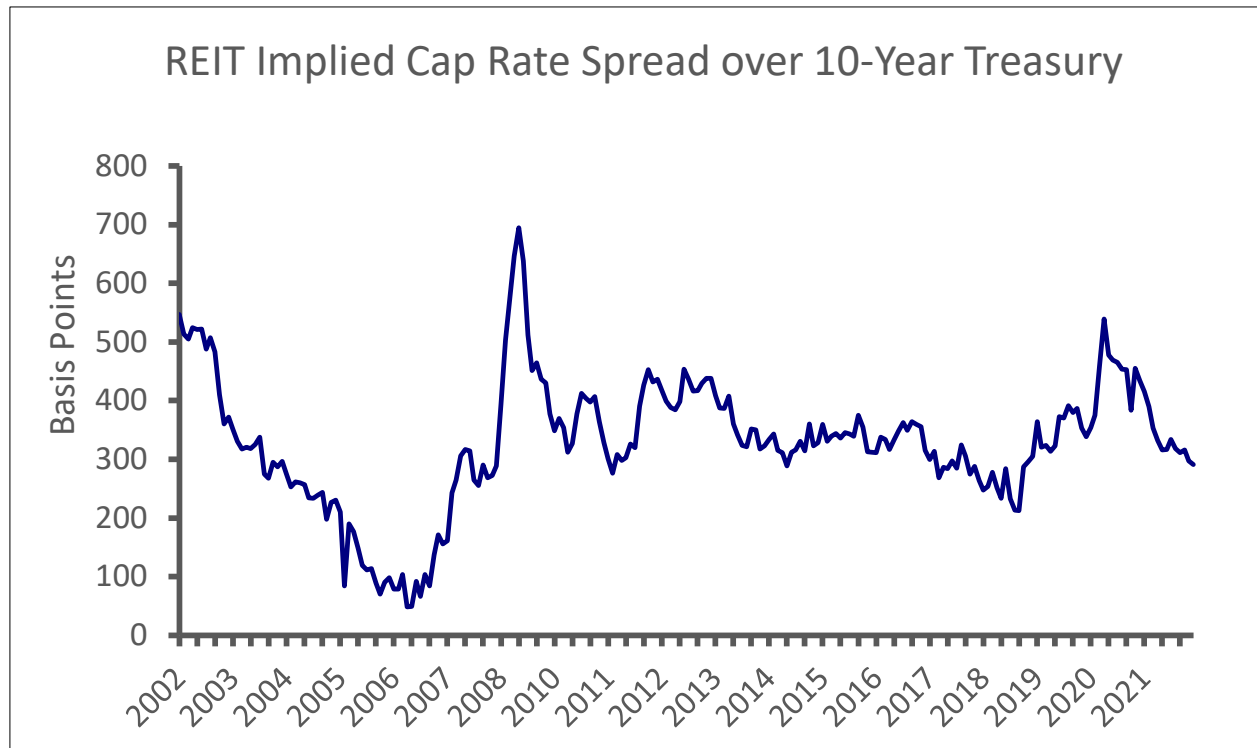


Figure 9.8

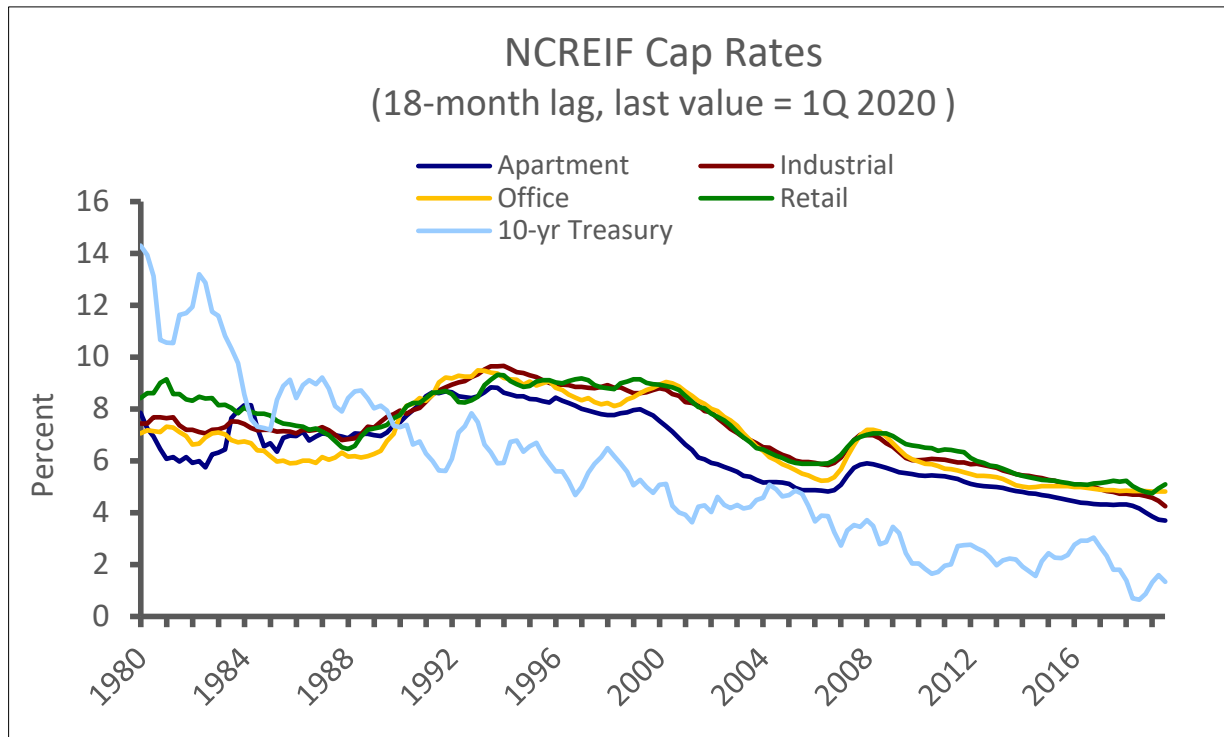


Figure 9.9

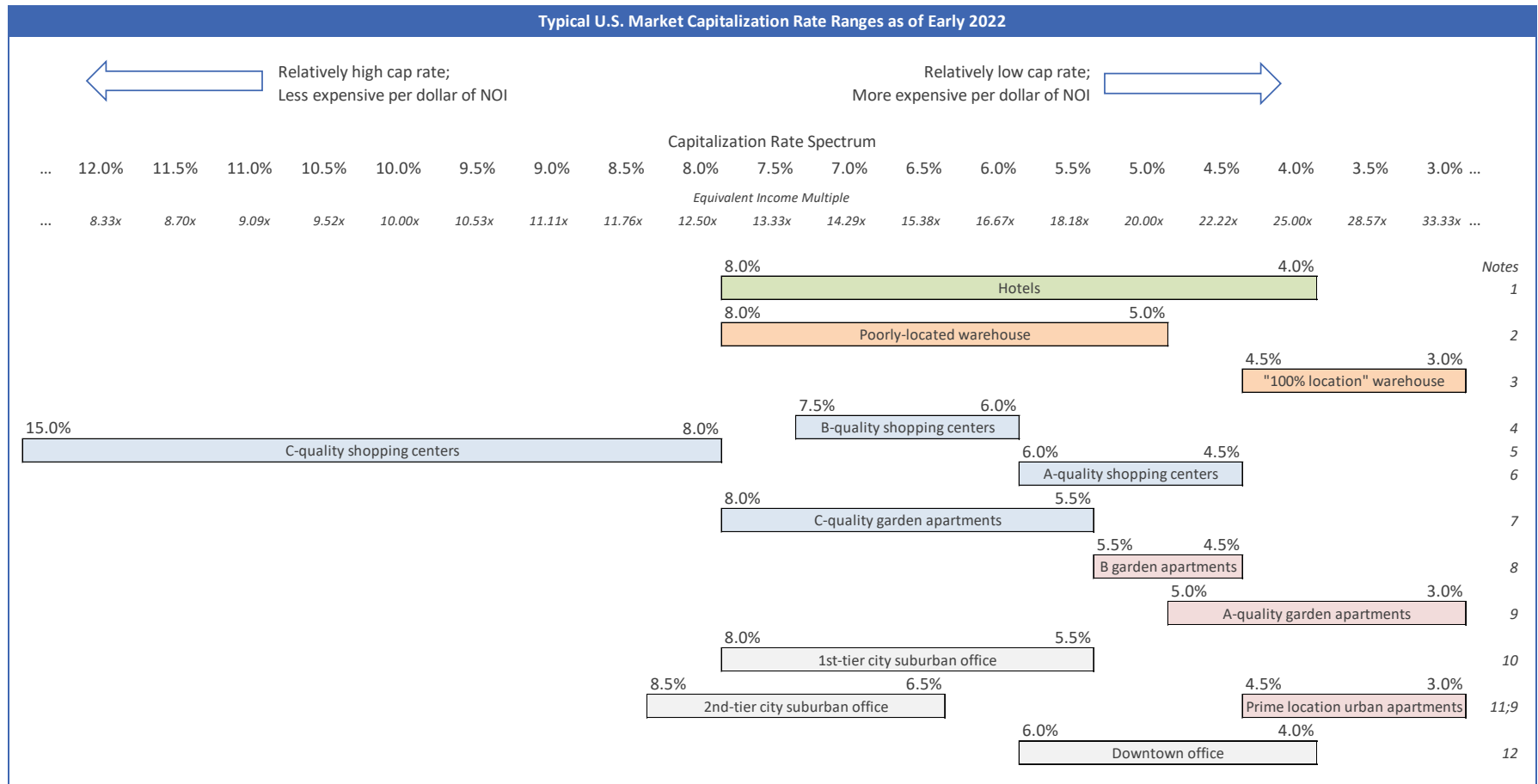


Figure 9.9 Notes

- 1 **Hotels:** Hotel properties are difficult to value due to their operating leverage. Hotels are in the 18-hour lease business, and they are essentially all fixed costs. Regardless of the occupancy from night to night, you are going to incur costs to continue to operate and maintain the hotel, the soap, the electricity, the marketing costs, etc. Therefore, a decrease in demand will cause a quick decline in prices. Hotels will be willing to lower prices to draw as many people as possible to cover as much of the fixed cost as possible. If you don't do not cut your prices, you will remain empty. If you do cut, you will have lower income. This is a good example where cap rates and multiples are inappropriate valuation metrics because of unstabilized income. This is also an example of the numbers serving as a starting point. You could perform a DCF analysis on a hotel property right now. However, you are really going to have to start thinking about the risks and opportunities beyond the DCF. There is no way the model will be able to accurately capture the possible outcome for the property.
- 2 **Poorly-located warehouse:** Poor location with no easy access to the highway. Warehouses tend to have higher cap rates than other property types. This is because warehouses tend to have lower growth rates. Growth rates are lower because you can build a warehouse in roughly 8 months. A warehouse is generally a fairly simple structure to build. Therefore, demand can rarely exceed supply by a great margin for an extended period of time. If rents start to increase, developers can quickly build new product that will satisfy the demand. Furthermore, there is additional risk with a warehouse property. Specifically, warehouses only rent to a few large tenants, GE, Walmart, etc. Although those companies have excellent credit, your profitability is completely concentrated with a few tenants. If the tenant decides to leave, you are faced with an empty building and no cash flow.
- 3 **"100% location warehouse":** near a major highway or exit ramp that leads to the property with ease.
- 4 **B-quality shopping centers:** pretty good retail strip centers.
- 5 **C-quality shopping centers:** empty Kmart, not well located.
- 6 **A-quality shopping centers:** good supermarket-anchored tenant with Wal-Mart already in the competing market, so no fear of sudden entry and decline in customers.
- 7 **C-quality garden apartments:** usually has a good market. There is a significant increase in the cap rate for lower quality, B and C, properties. In a tough market, the A quality properties will be able to maintain occupancy. Although owners may have to lower rents to keep tenants, the B and C properties will have roughly the same occupancy. At lower rents, more people will be able to afford the A quality property and will move into the best buildings. As more and more people move to newer or higher quality buildings, the lower quality buildings will experience an increase in vacancy rates. With this higher risk, investors have historically required a higher return on these lower quality buildings, thus a higher cap rate. There is no exact definition for these quality rankings. This is just a phraseology the market uses to describe certain properties relative to other properties. It is just like saying someone is short. There is no pure definition of short.
- 8 **B-quality garden apartments:** usually has a good market.
- 9 **A-quality garden apartments / prime location urban apartments:** good location, recent design, does not rely on individual tenant credit, but tenants as a whole have good credit.
- 10 **1st-tier city suburban office:** suburban, good quality, smaller, fewer tenants, not as well located, not quite as liquid
- 11 **2nd-tier city suburban office:** small, two tenants, medium quality credit tenants, less liquid, lumpy lease expiration. There is a large spread on the second-tier office buildings. The city in which the property is located will have a significant impact on the property's cap rate.
- 12 **Downtown office:** prime towers, well-located, well-designed, top of the market, big building, lots of quality tenants, smooth lease expiration.

Figure 9.10

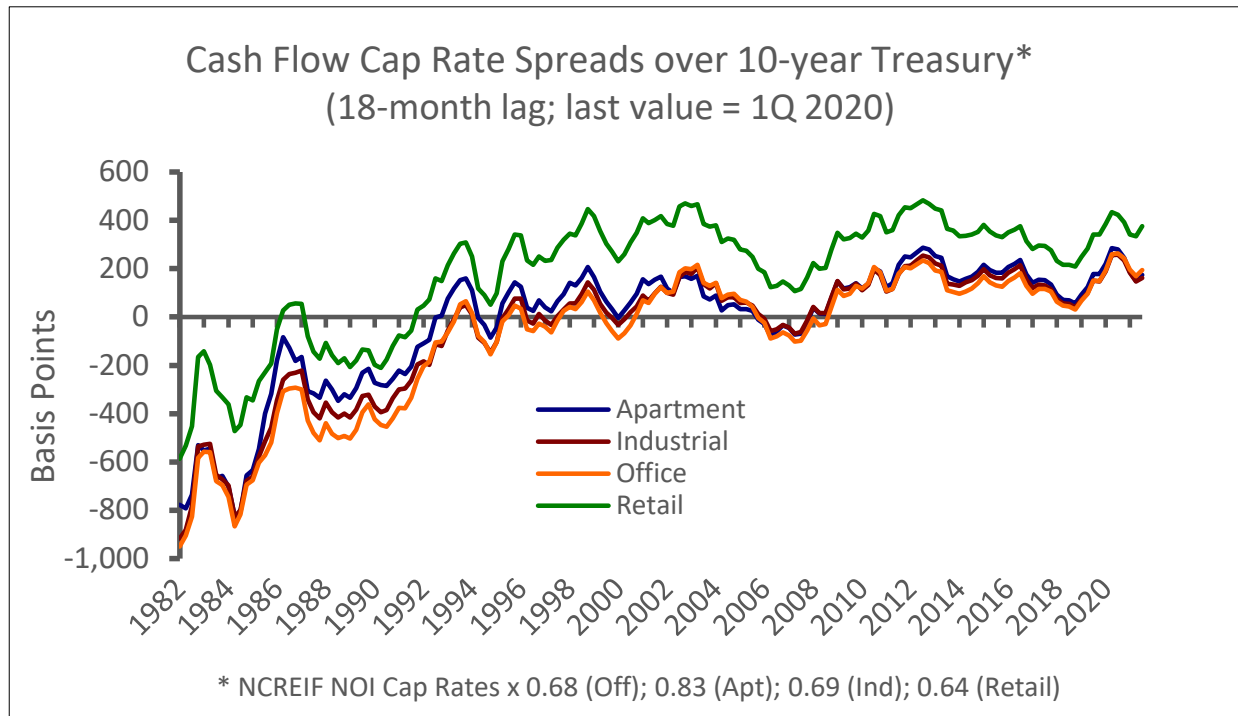
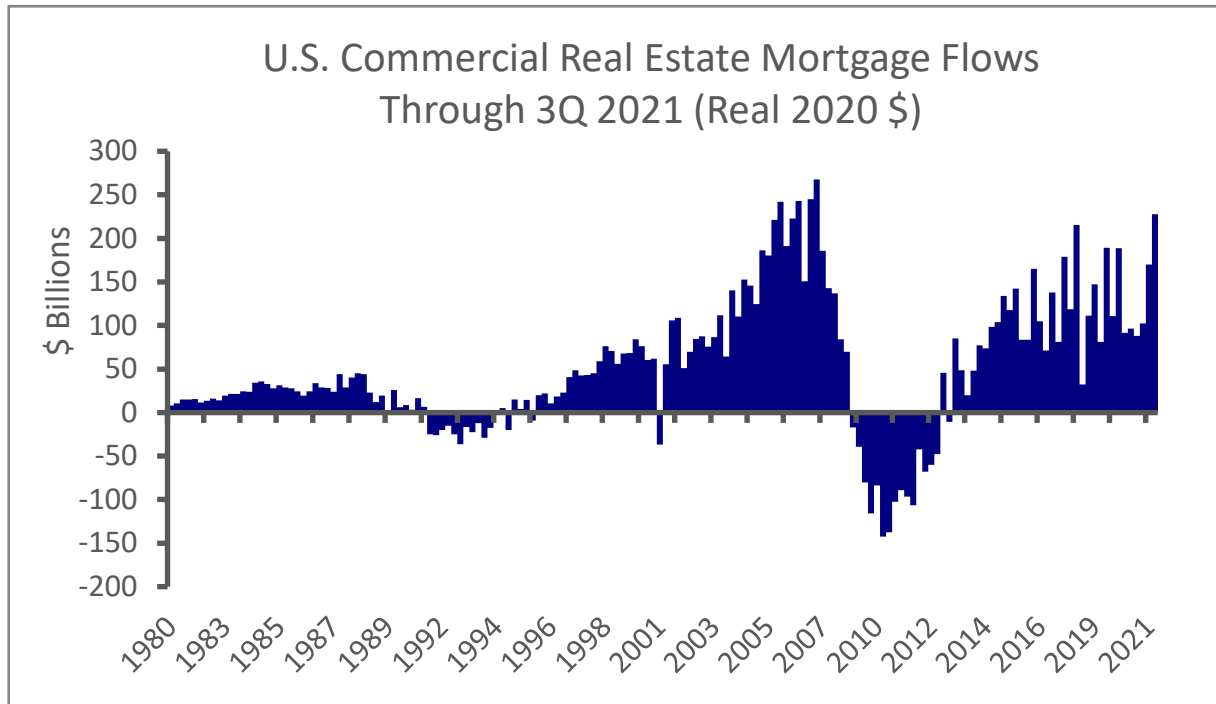


Figure 9.11

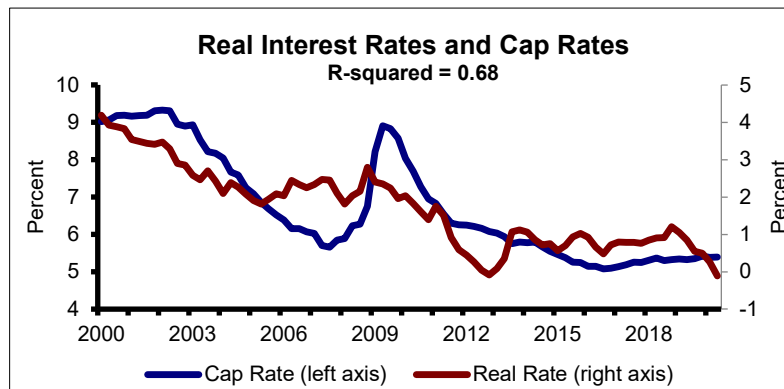
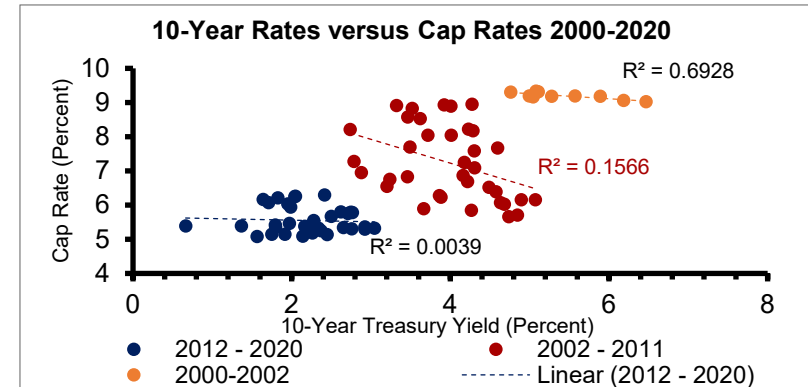
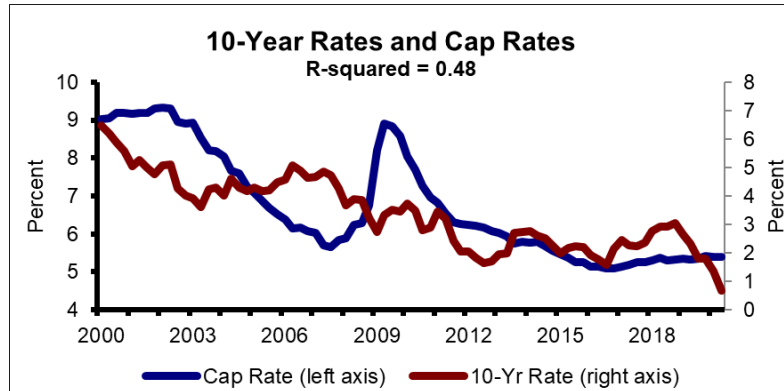


Note: negative points represent a combination of loans called and not renewed, loans where capital true ups were required, loans repaid on a sale with the new purchaser using less debt, and foreclosures where debt converts to equity.

What Determines Cap Rates More?

- Interest rate changes, or the availability of debt capital?
- Linneman Associates and Matt Larriva, CFA's research finds the latter is much more highly correlated to changes in cap rates
- The more mortgage capital available, the lower cap rates tend to go

What Determines Cap Rates More?



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